

Flu Forecasting Project DRAFT

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Abstract

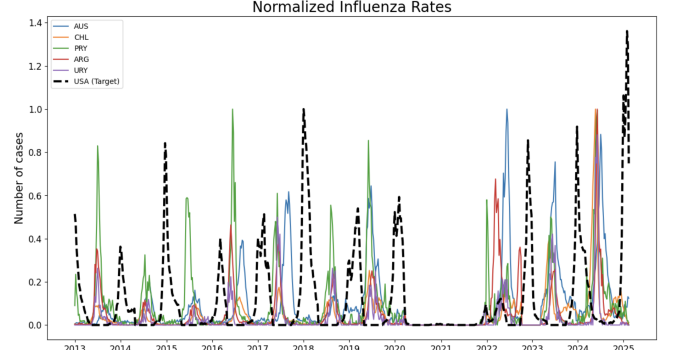
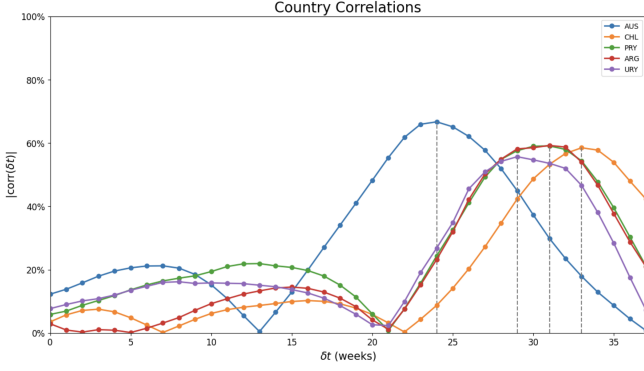
Each year, influenza epidemics result in significant morbidity, mortality, and cause adverse effects to the economy. Predicting annual influenza outbreaks can prepare jurisdictions to reduce detrimental outcomes through targeted vaccinations, health care preparations, and other programs to reduce exposure. Since probability of exposure among individuals is enhanced through increased human density, we used incidence data from the Southern hemisphere during the winter to predict future influenza rates in the Northern hemisphere. The model predicted a 30% increase in the winter of 2025 in North America which closely tracked actual incidence. We maintain models populated with data from other jurisdictions can be useful to predict relative incidence and be used to more effectively allocate resources from which to protect public health.

1 Introduction

Annual influenza outbreaks cause considerable morbidity, mortality, and economic effects. There is also widespread concern amongst public health professionals regarding the deadly potential of a future influenza pandemic (Doshi 2008). Pathogenic airborne viruses increase transmission proportionally within higher areas of human density. Such increases occur annually during winter in temperate regions when people spend greater time indoors (Simonsen 1999). Tempo-spatial patterns can be useful in predicting influenza outbreaks (Onozuka and Akihito 2008, Shaman and Karspeck 2012) and can be used in modeling approaches to better prepare certain jurisdictions to focus public health resources.

Long range influenza forecasts are scarce due to the high variability of the influenza season year-to-year. The current state-of-the-art models provided by the CDC (Flu-Site, find better citation and explanation of their work) are done using an ensemble that predict 1-4 weeks influenza hospitalization rates. Their work is mostly motivated to aid hospitals in preparation for potentially overwhelming flux of patients. Longer forecasts can be beneficial for the pharmaceutical manufacturers who need to scale production months in advance, and can be hesitant to make a surplus of products that will go unsold.

The strength of an influenza outbreak can be dependent on a variety of factors such as the dominant strain(s) of the virus $H(i)N(j)$, vaccination efficacy and adherence, timing of close human activity etc. Vaccine efficacy relies on the CDC's ability to predict the dominant viral strains likely to infect humans. Vaccines are created by analyzing the current dominant strains effecting humans around the globe to select "candidate vaccine viruses" (CVVs) to include in the new vaccine formulation where the mutation behaviors are studied. Vaccine efficacy varies year-to-year but has had a relatively small variance in the past 10 years with a 36(9)% estimated reduction in infection. Rather than studying the viral mutations, which has already shown to be very challenging as evident by vaccine efficacy never exceeding 50%, we propose countries in the southern hemisphere can act as confounding variables where the necessary viral properties and human behavior are hidden and manifest themselves in each country's respective severity rate.



(a) The magnitude of the correlations computed between southern hemisphere countries influenza case records with the strongest signal with the US influenza hospitalization rates. The correlations are with respect to the number of lagged previous weeks (δt). (b) The US hospitalizations (black, dashed) between 2015 and the beginning of 2025 compared to the 5 southern hemisphere countries total influenza cases documented by WHO. All countries are normalized to the most severe week that country experienced as documented by the WHO and the CDC.

This work aims to push the flu modeling forecasting range to approximately $\frac{1}{2}$ -year, sacrificing some week-to-week accuracy, but captures the spirit of the influenza season. We first present a subset of countries whose influenza rates may be heavily related to the U.S. Following a proper “candidate” determination and preprocessing of the data, we propose model architectures built from Recurrent Neural Networks (RNNs), that have the ability to *learn* temporal relationships of the data to forecast the U.S. season. Following model declaration, we explain the model *tuning* process and compare influenza forecasts for the 23-24 and 24-25 seasons.

2 Southern Hemisphere Countries

The nature of the Earth’s tilt, with respect to its orbit around the Sun, creates a phase difference for northern and southern hemisphere seasons, making winters $\sim \frac{1}{2}$ year apart. Given the ease people travel the globe, we assume the viral strains dominating the infections in the southern hemisphere will be related to the strains dominating the northern hemisphere and are not isolated in a way that would cause significant deviations in strains.

A country’s size and latitude will generally dictate the significance of season change, so not every country in the southern hemisphere will serve as an equal predictor of the U.S. season. We select a list of relevant “candidates” by borrowing influenza records from the W.H.O. (cite) and compute their lagged correlations with the U.S. hospitalizations records given by the CDC (specifically this is the flu-surv dataset which represents $\sim 10\%$ of the population)¹. Specifically, we compute

$$C_{\alpha}(\delta t) = \frac{\sum_t (x_t^{US} - \bar{x}^{US})(x_{t-\delta t}^{\alpha} - \bar{x}^{\alpha})}{\sqrt{\sum_t (x_t^{US} - \bar{x}^{US}) \sum_t (x_{t-\delta t}^{\alpha} - \bar{x}^{\alpha})}} \quad (1)$$

where x_t^{α} represents the number of positive cases reported for country α on week t . Australia, Chile, Paraguay, Argentina, and Uruguay have the clearest signal as evident by their peak structures shown in Figure 1a. These countries have clear peaks in the $\delta t = 20 - 35$ week lags making them suitable candidates for our half-year forecasting goal.

It is evident in Figure 1b that there is no single candidate or obvious linear combination of the group that could represent the target. Each candidate has its own peak height, location, and width. There may be a non-linear relationship that can be captured via tuning a neural network.

¹We could just as well have used WHO data as the target as well, but CDC hospitalizations were chosen to avoid bias from having the entire project built on a single data source.

3 Models

RNNs (Recurrent Neural Networks) were once the state of the art for sequence learning commonly applied in time-series forecasting and text sequence generation (find some citations). They have lost popularity with the emergence of transformer models (citation), but can still be preferable for small datasets with significant temporal dependancies.

We employ RNNs to utilize their feedback connections that enable them to learn sequences of data where ordering is important. These have the particular advantage of not needing a parametric description to capture the seasonality, which can be challenging for weekly inputs where the week start date and year start date do not coincide. The input features the 5 southern countries in batches of 26 week long sequences shifted by 21 weeks making the input tensor of shape $[T_{window}, 26, 5]$ where T_{window} is the total weeks included in the fitting procedure.

The model architecture can be separated into a RNN and a sequential portion visually depicted in Figure 2. The RNN piece features a single RNN or two RNNs with a linear layer acting as a feature transformation in between. This general architecture has been used in original text sequence generation techniques where the first RNN acts to capture the semantic meaning of the sentence, the linear layer transforms this semantic representation, and the second RNN is used to predict the next word. We borrow this technique to apply to weekly influenza rates where the first RNN captures the behavior the Southern Hemisphere flu behavior, linear layer transforms into a new representation where the second RNN can then apply to predict the United States flu behavior. We make use of three flavors of RNNs by including Gated Rectangular Units (GRUs) and Long Short Term Models (LSTMs) to add additional variability to our model construction in an effort to not bias our forecast via a single model prediction. The sequential layers are used in dimensionality reduction with the hope they capture the remaining behavior via linear layers, RELUs and drop out layers when multiple linear layers are used.² The various layer archetypes are illustrated in Figure 3.

We omit seasons 20-21 and 21-22 due to the damaging effect Covid-19 had on the seasonality. The input during that time period were replaced with zeros to not damage the continuity of the dataset, but we acknowledge we could have just as easily removed that period entirely from training.

We attempt to predict an entire season from previous seasons' data in this so it as natural to use a full season as validation when training the model in an effort to minimize the inevitable overfitting that neural networks are guilty of. Using weekly data has an advantage over monthly by increasing the sample size, and it is a more natural stratum of influenza records given the rapid change in the season, but does not increase it large enough to avoid overfitting alone. We utilize early stopping and drop out layers in the sequential piece in an effort to combat the overfitting nature of neural networks.

We used mean squared error as the loss function for back-propagation, however invented a slight hybrid to the conventional mean squared error function to evaluate the validation set for early

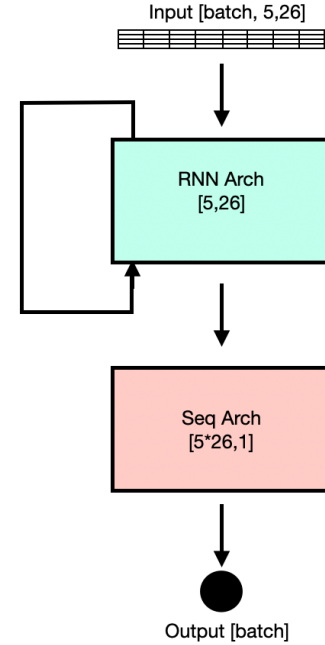


Figure 2: Model archetype composed of two varying combinations of RNNs and sequential layers.

²Did not want to use a dropout layer from the RNN output due to the fact that all of that temporal information is important

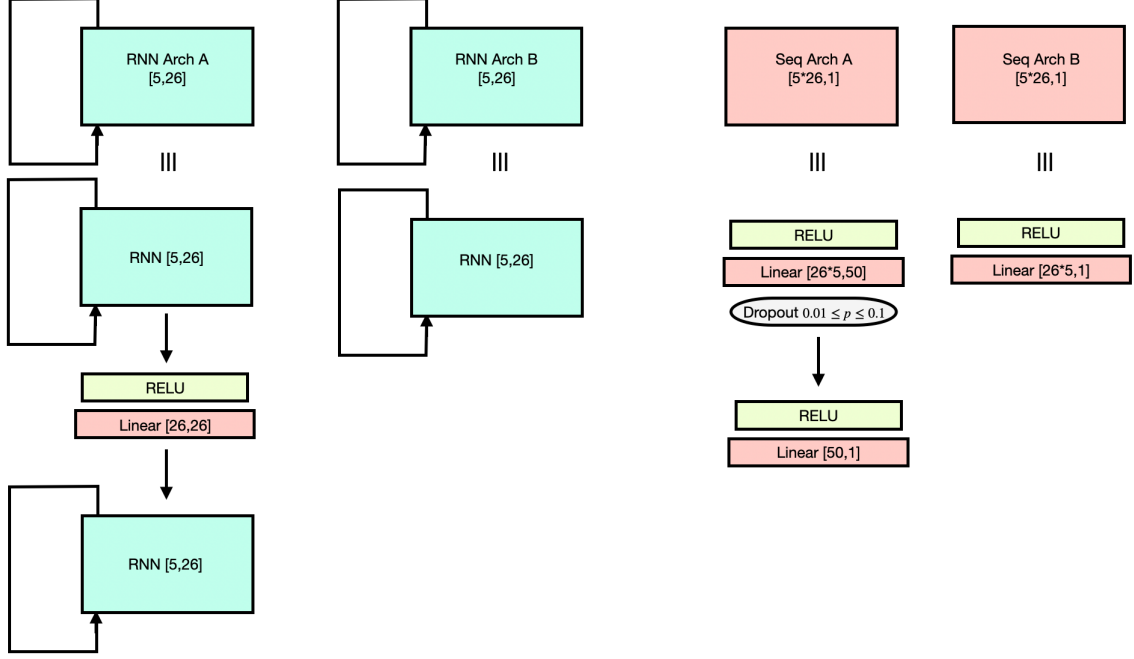


Figure 3: Model archetype composed of two varying combinations of RNNs and sequential layers.

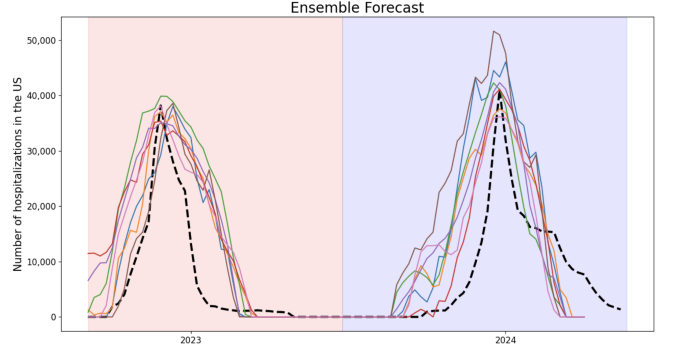
stopping. Our validation metric is the following

$$MSE^{adj} = \left(y_{val}^{max} - y_{val}^{pred,max} \right)^2 + \alpha^2 \left(\left(t_{val}^{max} - t_{val}^{pred,max} \right)^2 + MSE_{train} \right) \quad (2)$$

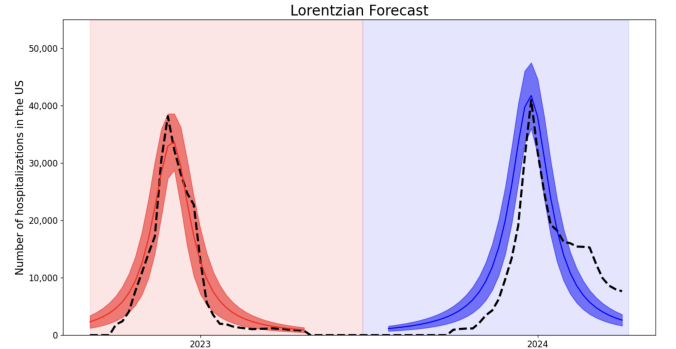
where y^{max} is the peak height, t^{max} is the peak location, and the MSE_{train} is the mean squared error of the training set. This metric ensures the loss function is not totally penalized for not capturing the exact week-to-week behavior, but can still capture maximum weekly rate and location. The training mean squared error is included to ensure the model represents the training set, but is suppressed by a factor of $\alpha = 0.1$ so that the peak height is still the dominant feature when searching for a minima or plateau in the training curve. We suppress the peak location difference as well to keep it on the same relative scale as the peak height.

We found $MSE^{adj} \sim 0.2$ adequately described the validation peaks while simultaneously allowing the model to describe the training set. The accepted model variations with $\min(MSE^{adj}) \leq 0.2$ are displayed in Figure 5. These models capture the peak height and location but possess spurious movement outside of the flu season and we remove these from the final forecasts.

A naive weekly average of the model variation will produce a poor description of the data due to the seasonality where a any small phase difference of two otherwise completely equivalent descriptions would wash



(a) Model variations validated in 22-23 season (red) for 23-24 forecast (blue).



(b) Single-peaked parameterization for validation and test set with parameters means and standard deviations are used to create model parameterization's mean and uncertainty.

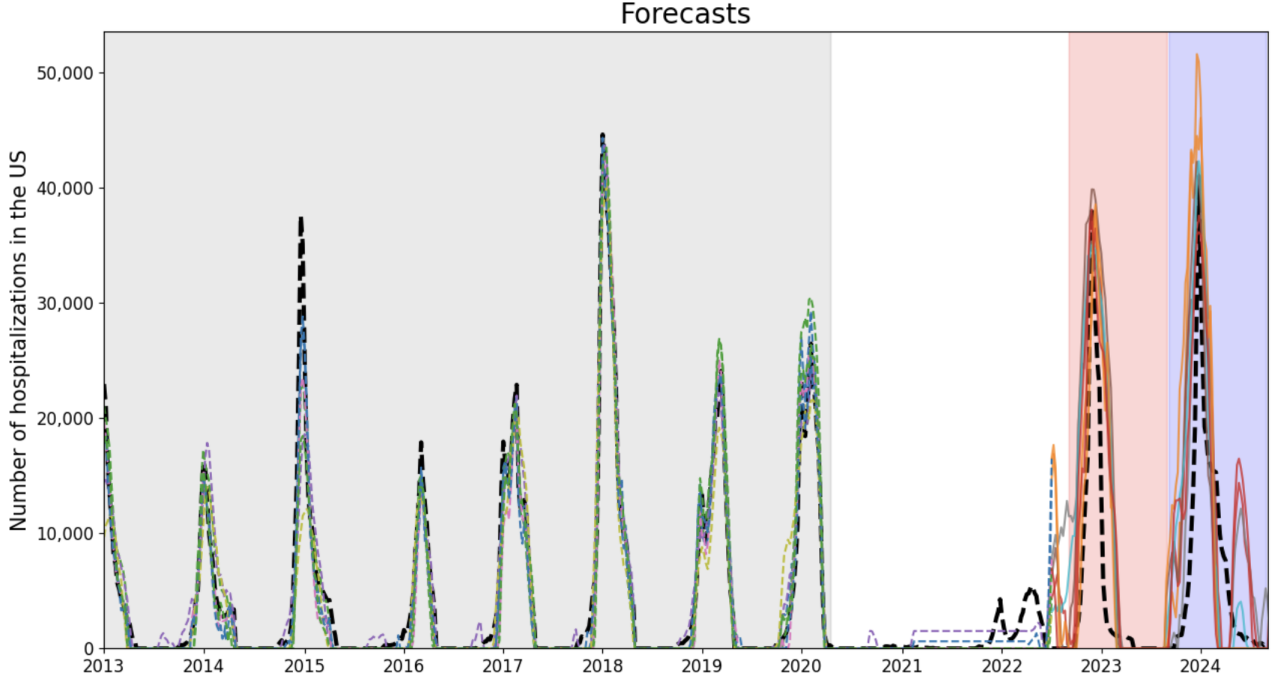


Figure 5: Dashed line represents the target U.S. hospitalization rate. Different colors represent model variations appear as dashed in the training region and solid in the validation and test region. Validation and test regions are represented with red and blue shaded backgrounds respectively.

away the peak behavior. One must first capture the phase differences and shift accordingly. A parametric solution to combining is the following sum of renormalized hybrid Lorentzian distribution

$$L(t) = \sum_i^2 A_i \frac{\left(\frac{\Gamma_i}{2}\right)^2}{(t - t_i)^2 + \left(\frac{\Gamma_i}{2}\right)^2} \quad (3)$$

where A_i is the peak height, t_i peak location, and Γ_i is the half width. A typical Lorentzian has a defined normalization that we have removed to control the height. All three of the parameters can be computed for each model and their averages and standard deviations can be used to estimate the parameterization variance.

Each season appears to have one or two-pronounced peaks meaning we can reduce the ~ 30 -week flu severity rates to three or six parameters. We demonstrate this relationship in Figure 7 where we have used two peak models to fit each season in the past 10 years with mean absolute error varying from 1.6% to 4.0%.

Our models may be capable of capturing the peak height, but cannot distinguish exact timing or whether the season is single or double-peaked. It may be tempting to use the fit parameters as targets and reduce the dimensionality of the problem, however training becomes challenging because each year is reduced to a single data point. As a reasonable simplification, we carry forward using single-peaked approximations of seasons where the Lorentzian parameters are well defined from model predictions.

3.1 23-24

We show the results of the model variability and parameterization side-by-side in Figure's 4a and 4b respectively. Peak height and location are simple to interpret from the model results while

the half-width is slightly more challenging to measure. This was computed as an average of the half-width defined via the ascent and descent of each individual model.

The model produces a peak height of $41.8K(57)$ patients compared to the actual peak height of $40.0K$ patients with a total case count of $406K(104)$ patients to the 23-24 total of $367K$. The peak height maintains $\sim 13\%$ uncertainty with the total case count having $\sim 25\%$ uncertainty. The model remarkably predicts the peak height accurately with reasonable precision. Most of the uncertainty in the model lies in the duration of the season which is unsurprising given that the season durations vary year-to-year and the data input will have no knowledge of that feature.

3.2 24-25

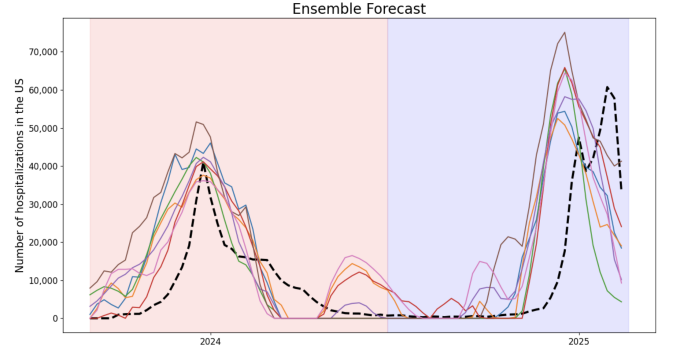
We repeat the same procedure for the following season and show the model results in Figures 6a and 6b. In this season we have two peaks and we remind the reader we make only an effort to model a single peaked season. We choose to compare to the dominant peak as we did in the previous season. Our model produces a peak height of $60.5K(86)$ patients compared to the actual height of $60.7K$ patients. Our peak height has $\sim 14\%$ uncertainty similar to the previous season, maintaining a relatively accurate and precise dominant peak height, but naturally fails to describe the double peaked nature of the season due to our approximation.

4 Conclusion

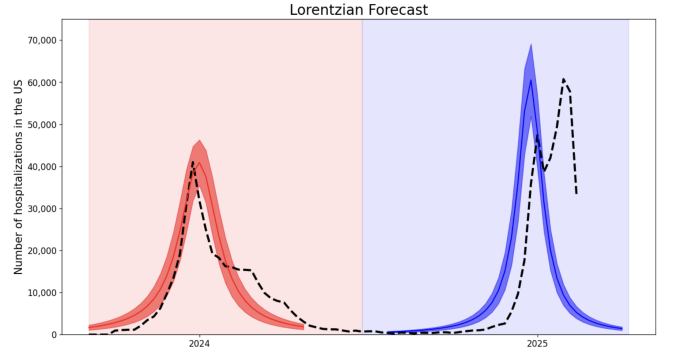
The results of the past two seasons are promising first steps in long-range influenza forecasting. We demonstrated that southern hemisphere countries can serve as leading indicators of the U.S. influenza season and showed that the 30-week flu season can be reduced to three or six parameters depending on whether the season has one or two peaks respectively. The parameters for a single peaked season can be uniquely determined by training a series of RNN architectures on a subset of the southern hemisphere influenza records. The models are able to describe the maximum hospitalization rate with high accuracy and precision, and shows potential of capturing peak locations within ~ 2 weeks.

As evident in the 24-25 season test case, further work is needed to describe multi-peaked seasons. There was a partial success in determining the multi-peaked season in that the maximum rate was well determined, however the single-peak approximation in this case was clearly not valid. We previously showed that the all seasons can generally be described as two peak seasons, so our future work will be targeting the forecasting of the lesser prominent peak as opposed to the dominant peak. Provided one can predict the second peak and their separation with reasonable accuracy, then the entire flu season can be determined up to some overall translational variance.

We acknowledge the Flu-Surv dataset as well as the W.H.O. for maintaining publicly available datasets used as inputs and targets for this model.



(a) Model variations validated in 23-24 season (red) for 24-25 forecast (blue).



(b) Single-peaked parameterization for validation and test set with parameters means and standard deviations are used to create model parameterization's mean and uncertainty.

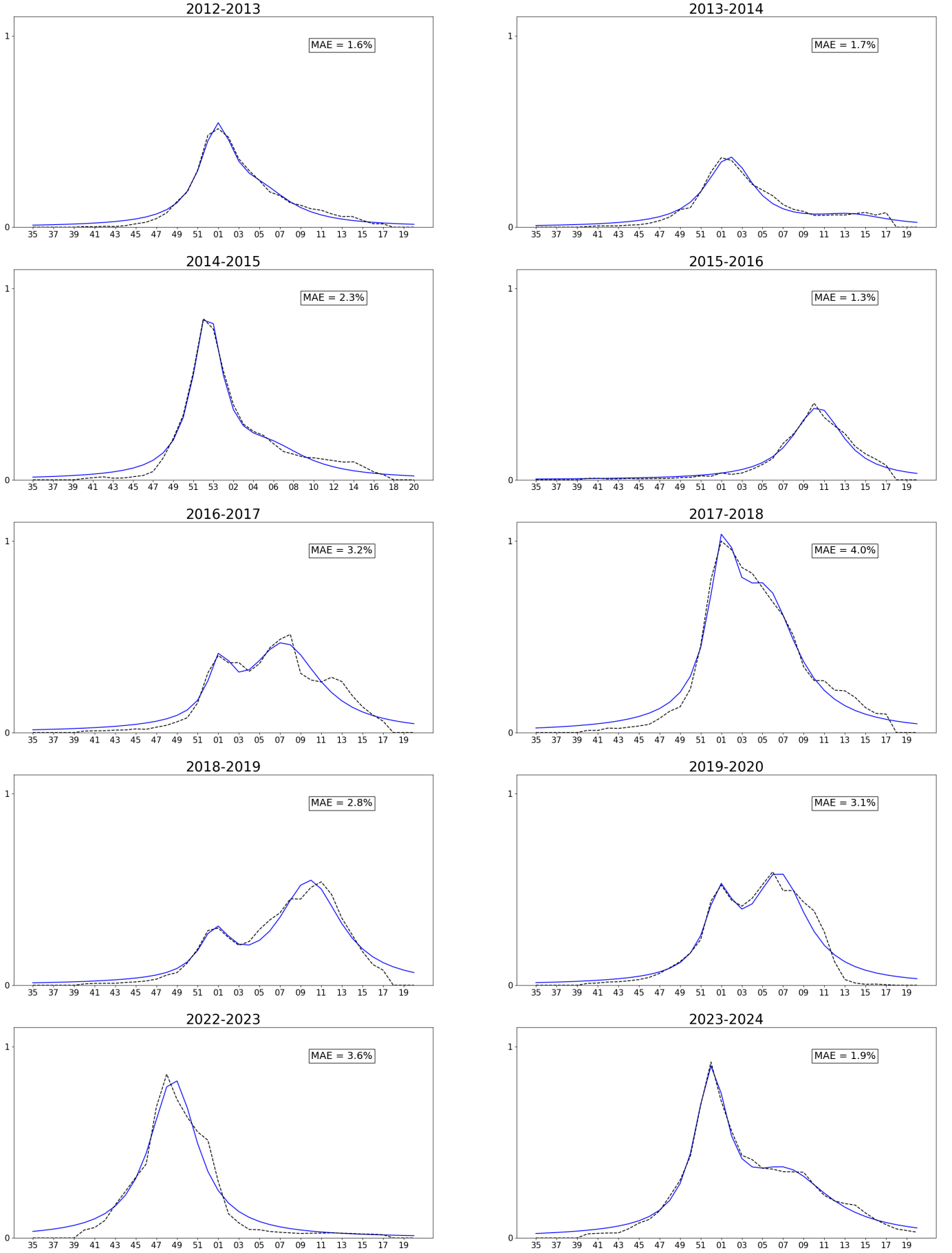


Figure 7: Six parameter determinations (see equation 3) of the past 10 significant influenza seasons. These descriptions show a mean absolute error varying between 1.3% – 4.0%.